calm_rank_levered

equity strategy | family: equity-volatility | status: research | derived from `calm_rank_tilt`

What this strategy does

Calm Rank Tilt with its lower risk spent on larger bets (about 1.7x) and a 4x borrowing limit

Why it should work. PEAD alpha per dollar is highest in calm stocks; rank-weight toward them without depending on vol levels.

Design notes (from the strategy's source file).

Calm Rank Tilt -- Balanced Leverage, tilted hard toward the calmest stocks by volatility RANK (research).

Calm Weighted sized each event by 1/idiosyncratic-vol (Sharpe 1.30 -> 1.47). The attribution work showed that PEAD alpha per dollar is concentrated in low-volatility names and almost absent in the wildest tercile, and that the 1/vol weight, being linear in the vol LEVEL, only mildly demotes the high-vol tail. Weighting by (1.02 - vol percentile)^2 -- where the percentile is measured against the previous quarter's cross-section within the stock's size quintile -- demotes it hard while staying smooth:

Sharpe 1.30 (Balanced Leverage) -> 1.47 (Calm Weighted) -> 1.81; max drawdown -16.6% -> -11.6%; every era improves split-sample check: exponent picked on 1996-2004 data alone (best k ~ 3) lifts 2005-2013 Sharpe 0.93 -> 1.10, the same as the full-sample best, so the gain is not an artifact of one period.

Lower volatility buys room: `calm_rank_levered` spends it on 1.7x the base unit and a 4x gross cap for ~9% a year at ~5% vol (a market-neutral book clearing the S&P 500's return in this sample). More sizing than that stops paying: the 5%-of-ADV liquidity cap starts to bind and 2008-13 Sharpe decays.

STATUS research: the exponent and sizing were chosen after seeing Calm Weighted's full-sample results (the split-sample check above is the only out-of-sample evidence), and the levered variant leans on small-cap capacity.

parameter	value
base_unit_fraction	0.005
max_gross_leverage	4.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	calm rank levered	market (CRSP VW)
Annualized return	8.07%	9.94%
Annualized volatility	5.70%	19.35%
Sharpe (NAV-pct)	1.39	0.59
Sortino	2.08	

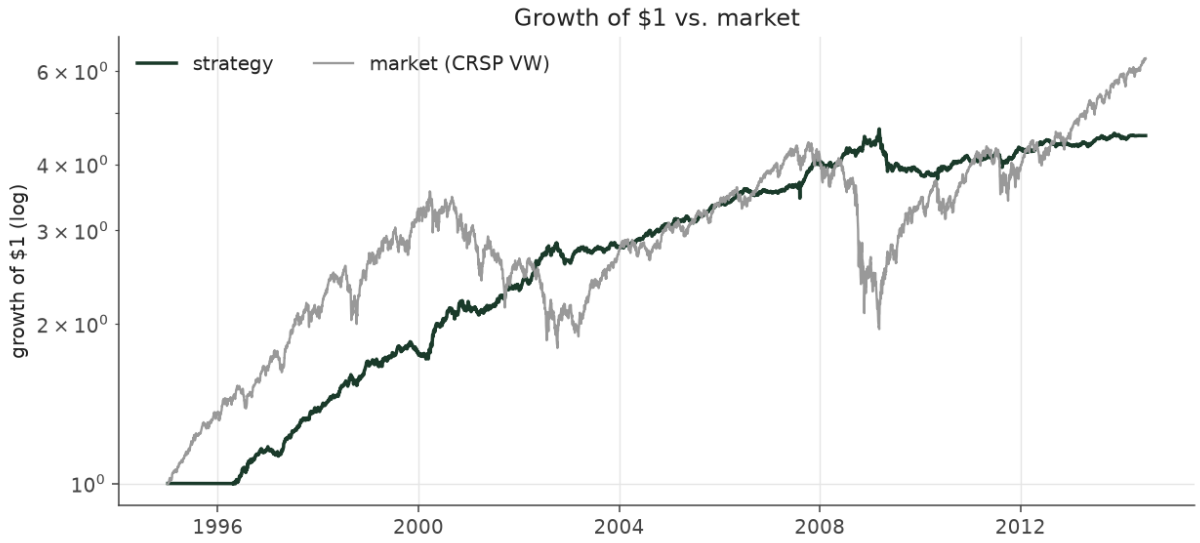
Max drawdown	-19.6%	-55.5%
Calmar	0.41	
Final NAV ($10M start)	$45,295,760	
Worst day	-4.00%	
% positive days	51.4%	
Sharpe 1996-2001	2.17	
Sharpe 2002-2007	1.66	
Sharpe 2008-2013	0.38	

Correlation of daily returns with the market: **-0.10**.

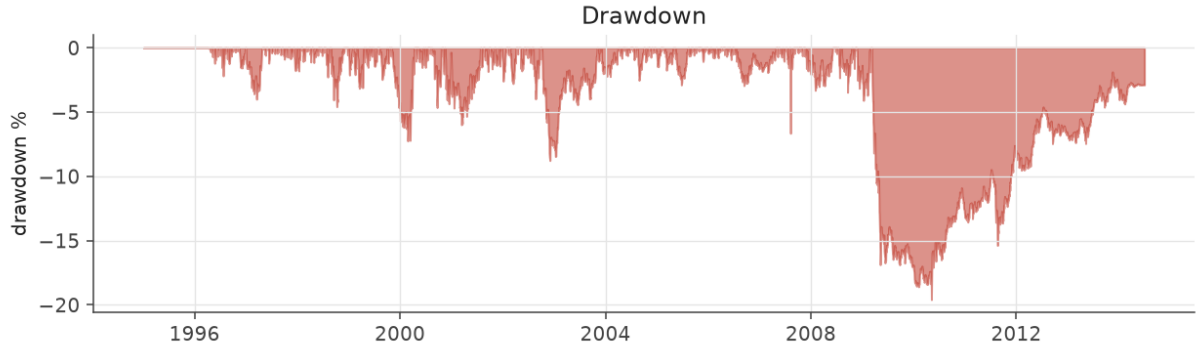
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 1.39

• best return among equity: `market_boost_max` (0.15); this one: 0.0807

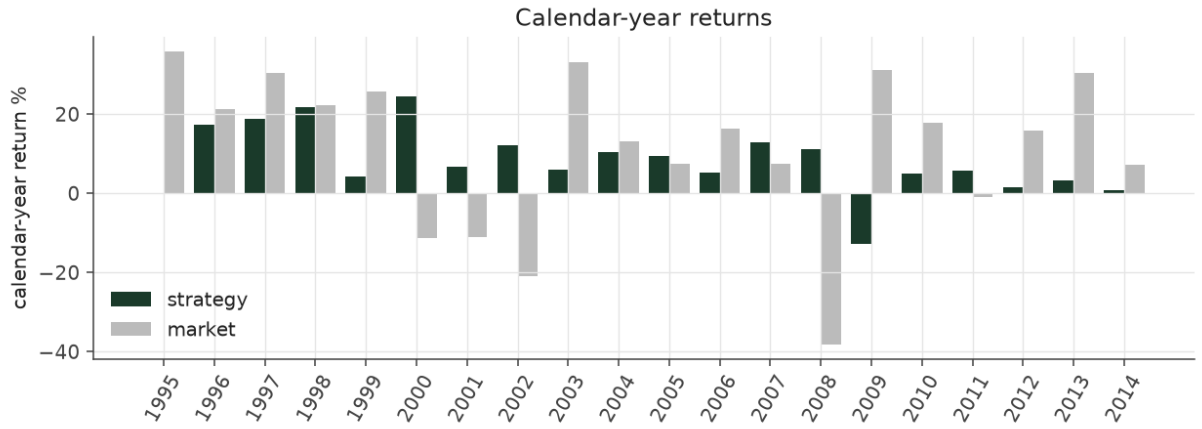
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.412



NAV (log scale)



Drawdown from running peak

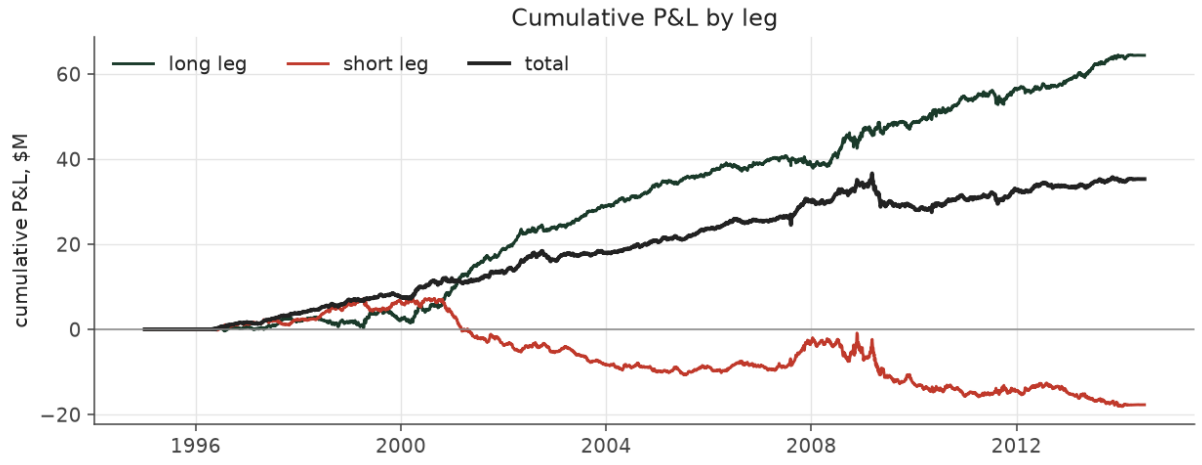


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	64.36	-17.72
P&L after costs ($M)	58.81	-23.51
of which common (average-event) component ($M)	47.25	-47.58

of which excess / PEAD component ($M)	17.11	29.86
excess minus costs ($M)	11.56	24.07
gross return per $ traded (bp)	213.15	-58.45
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.31	-0.26

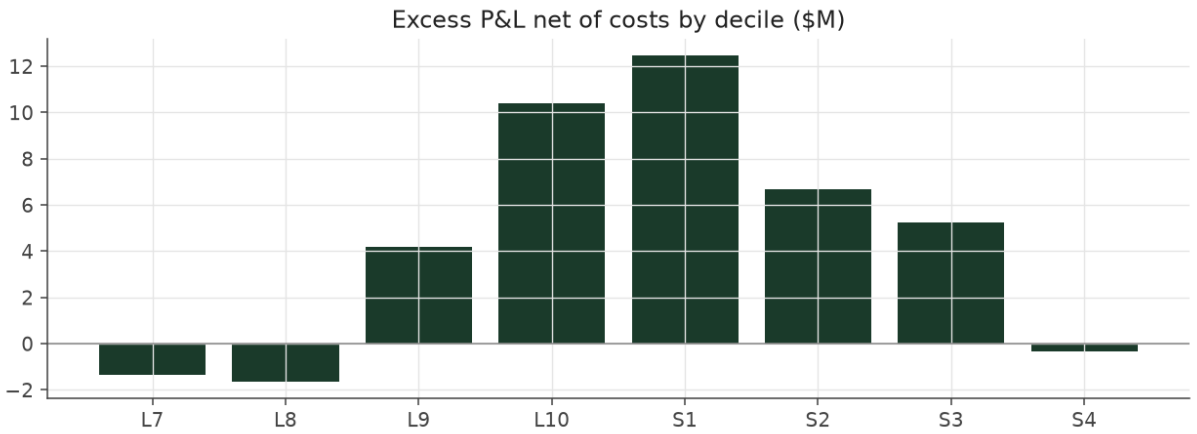
Daily P&L correlation between the two legs: **-0.76**.

• **The short leg loses money in absolute terms** (-17.7M before costs) while the long leg earns +64.4M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 47.2M / 47.6M. Stripping that common component, the surprise signal itself earned +17.1M on the long side and +29.9M on the short side.

• The two legs' daily P&L are -0.76 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.31 long / -0.26 short vs 1.39 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	6,446	221.13	2.31	-0.93	0.41	-1.35	-3.8%	-60.86
long	8	16,531	691.72	10.13	-0.38	1.28	-1.66	-4.7%	-23.99
long	9	18,481	922.83	17.24	5.86	1.68	4.18	11.7%	45.33
long	10	20,051	1,183.81	29.13	12.56	2.18	10.38	29.1%	87.72
short	1	18,225	1,123.91	-5.86	14.69	2.22	12.47	35.0%	110.92
short	2	18,826	1,044.37	-9.82	8.64	1.98	6.66	18.7%	63.81
short	3	15,255	771.21	-6.14	6.69	1.43	5.25	14.7%	68.12
short	4	2,350	92.05	-1.68	-0.16	0.16	-0.32	-0.9%	-34.75



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,938	327.96	1.25	3.34	1.24	2.10	5.9%	64.01
2	22,110	921.09	7.59	10.53	2.58	7.96	22.3%	86.37

3	24,900	1,352.22	9.92	12.58	2.87	9.71	27.2%	71.80
4	27,063	1,661.74	10.57	12.91	2.67	10.24	28.7%	61.62
5	28,154	1,788.03	5.97	7.61	1.99	5.62	15.8%	31.46

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,319	241.88	2.91	4.50	1.21	3.29	9.2%	136.05
2	22,452	1,045.90	6.13	10.78	3.14	7.64	21.4%	73.05
3	25,189	1,577.10	12.71	16.25	3.15	13.09	36.8%	83.02
4	27,001	1,638.51	8.36	9.64	2.29	7.34	20.6%	44.81
5	28,199	1,547.42	5.14	5.75	1.55	4.20	11.8%	27.15

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	41,550	2,201.76	1.87	18.24	3.92	14.32	40.2%	65.03
41-60d	27,518	1,482.19	7.72	13.57	3.06	10.51	29.5%	70.94
61-90d	37,656	1,971.30	18.32	8.68	3.34	5.34	15.0%	27.09
>90d	9,441	395.78	7.39	6.48	1.02	5.45	15.3%	137.81

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,241	1,019.95	14.16	16.11	1.89	14.22	39.9%	139.41
2002-07	38,279	2,111.49	15.57	18.70	3.89	14.82	41.6%	70.17
2008-13	45,645	2,919.59	5.57	12.15	5.56	6.59	18.5%	22.58

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,022	622.83	3.95	4.88	1.26	3.61	10.1%	58.03
Chems	2,545	383.62	2.15	2.76	0.61	2.15	6.0%	55.94
Durbl	2,819	403.03	1.16	1.95	0.67	1.28	3.6%	31.80
Enrgy	6,318	469.94	4.90	5.93	0.81	5.12	14.4%	108.86
Hlth	11,333	581.58	1.65	2.52	1.12	1.40	3.9%	24.03
Manuf	11,872	563.11	2.46	3.83	1.09	2.75	7.7%	48.79
Money	17,196	562.65	4.35	5.72	1.19	4.54	12.7%	80.60
NoDur	4,899	470.37	3.02	3.72	0.82	2.90	8.1%	61.65
Other	17,930	604.27	4.83	6.22	1.28	4.94	13.9%	81.79
Shops	11,548	575.54	3.86	4.83	1.12	3.72	10.4%	64.57

Telcm	3,947	415.79	2.11	2.80	0.69	2.11	5.9%	50.81
Utils	4,736	398.28	0.87	1.80	0.69	1.12	3.1%	28.07

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	38,501	537.62	2.32	3.83	0.97	2.86	8.0%	53.26
low vol	38,978	3,340.68	20.09	27.40	6.38	21.02	59.0%	62.92
mid vol	36,897	2,077.78	11.70	14.53	3.81	10.72	30.1%	51.62

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	8.07%	-0.00%
0.5%	3.0	7.69%	0.38%
1.0%	5.9	7.29%	0.78%
2.0%	11.9	6.39%	1.67%
5.0%	29.7	2.31%	5.75%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.